

AIM : Decision Tree Learning

1. Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
2. Evaluate the accuracy and effectiveness of the decision tree on test data.
3. Visualize and interpret the generated decision tree.

CODE :

```
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report

# Load built-in Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

feature_names = iris.feature_names
class_names = iris.target_names

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Create Decision Tree
model = DecisionTreeClassifier(
    criterion="entropy",
    max_depth=3,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
```

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```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Dataset Shape:", X.shape)
```

```
print("\nActual Values:")
```

```
print(y_test)
```

```
print("\nPredicted Values:")
```

```
print(y_pred)
```

```
print("\nAccuracy:", accuracy * 100, "%")
```

```
# Classification Report
```

```
print("\nClassification Report:")
```

```
print(
```

```
    classification_report(
```

```
        y_test,
```

```
        y_pred,
```

```
        target_names=class_names
```

```
    )
```

```
)
```

```
# -----
```

```
# Decision Tree Visualization
```

```
# -----
```

```
plt.figure(figsize=(18, 10))
```

```
plot_tree(
```

```
    model,
```

```
    feature_names=feature_names,
```

```
    class_names=class_names,
```

```
    filled=True,
```

```
    rounded=True,
```

```
    fontsize=11
```

```
)
```

```
plt.title(
```

```
    "Decision Tree - Iris Dataset",
```

```
    fontsize=20,
```

```
    pad=20
```

```
)
```

```
plt.tight_layout()
```

```
plt.show()
```

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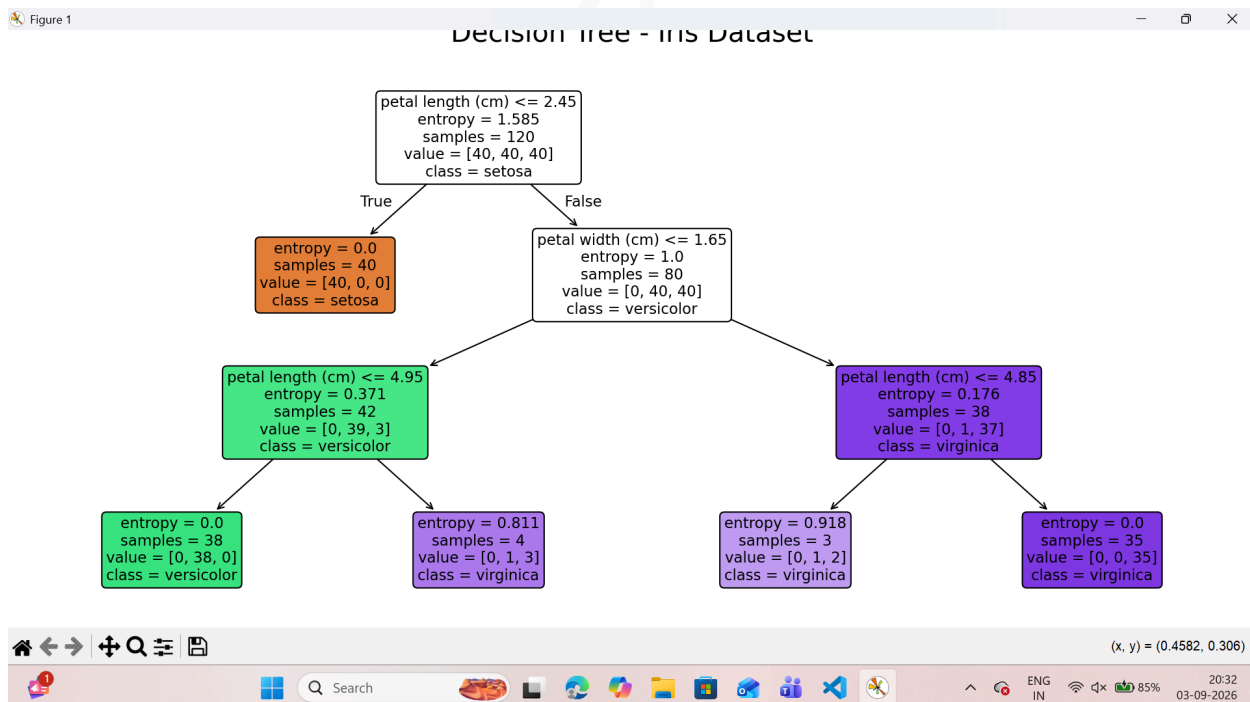
OUTPUT :

```

14 feature_names = iris.feature_names
15 class_names = iris.target_names
16
Problems Output Debug Console Terminal Ports
PS C:\Users\DELL\OneDrive\Desktop\AI Practicals> python Practical3.py
weighted avg 0.93 0.93 0.93 30
PS C:\Users\DELL\OneDrive\Desktop\AI Practicals> python Practical3.py
Dataset Shape: (150, 4)
Actual Values:
[0 2 1 1 0 1 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]
Predicted Values:
[0 2 1 1 0 1 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]
Accuracy: 96.6666666666667 %
Classification Report:
precision recall f1-score support
setosa 1.00 1.00 1.00 10
versicolor 1.00 0.90 0.95 10
virginica 0.91 1.00 0.95 10
accuracy 0.97 0.97 0.97 30
macro avg 0.97 0.97 0.97 30
weighted avg 0.97 0.97 0.97 30

```

Decision Tree :



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